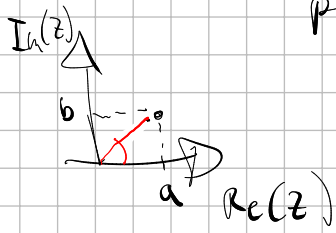


$$z \in \mathbb{C} \quad z = a + ib \quad i^2 = -1 \quad a, b \in \mathbb{R}$$

• modulo

polar form $\rightarrow z$ as vector $\rightarrow z = e(\cos \theta + i \sin \theta) = e e^{i\theta}$



$$a = e \cos \theta$$

$$b = e \sin \theta$$

$$e = \sqrt{a^2 + b^2}$$

$$\theta = \arccos \frac{a}{e}$$

$$\theta = \arcsin \frac{b}{e}$$

• called phase

$$z_1 \cdot z_2 = e_1 e_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) = e_1 e^{i\theta_1} \cdot e_2 e^{i\theta_2}$$

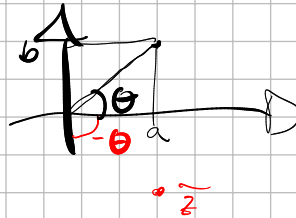
$$\frac{z_1}{z_2} = \frac{e_1}{e_2} (\dots)$$

$$z^n = e^n (\cos n\theta + i \sin n\theta)$$

$$|z| = \sqrt{a^2 + b^2} = \sqrt{z \cdot \bar{z}}$$

$$\bar{z} = a - ib = e(\cos \theta - i \sin \theta)$$

$$z^* = e e^{-i\theta}$$



$$e^z: \mathbb{C} \rightarrow \mathbb{C} \quad z \in \mathbb{C}$$

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$$

$$e^{a+ib} = e^a \cdot e^{ib}$$

$$e^{ix} = \sum_{k=0}^{\infty} \frac{(ix)^k}{k!} = 1 + \frac{ix}{1} + \frac{i^2 x^2}{2!} + \dots$$

$$= 1 + \frac{i^2 x^2}{2!} + \frac{i^4 x^4}{4!} + i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$\underbrace{\hspace{10em}}_{\text{even terms}}$$

$$\underbrace{\hspace{10em}}_{\text{odd terms}}$$

$$= \underbrace{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots}_{\cos x} + i \underbrace{\left(x - \frac{x^3}{3!} + \dots \right)}_{\sin x}$$

$$\cos x$$

$$\sin x$$

$$-e^{z+2\pi i} = e^z$$

$\forall k \in \mathbb{Z}$ periodic func

$$|e^z| = e^{\operatorname{Re}(z)}$$

$$-e^{-z} = \frac{1}{e^z}$$

$$-e^z = e^{\bar{z}}$$

$$-e^z \neq 0$$

$$z^n - 1 = 0 \quad n=2 \Rightarrow z^2 = 1 \Rightarrow z = \pm 1$$

$$z^4 - 1 = 0 \quad \text{roots } \pm 1, \pm i$$

$$z = \rho (\cos \theta + i \sin \theta)$$

$$z^n = \rho^n (\cos n\theta + i \sin n\theta) = 1$$

$$\rho = 1$$

$$\sin n\theta = 0$$

$$\cos n\theta = 1$$

$$n\theta = 2\pi k \quad k \in \mathbb{Z}$$

$$\theta = \frac{2\pi}{n} k$$

principal n-th roots of 1?

$$z_k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}, \quad k = 0, \dots, n-1$$

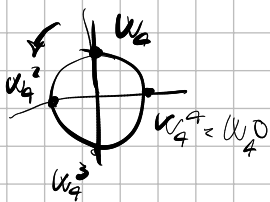
$$w_n = e^{\frac{2\pi i}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

$$w_4 = \cos \frac{2\pi}{4} + i \sin \frac{2\pi}{4} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$$

$$w_4^2 = \cos \frac{2\pi}{4} \cdot 2 + i \sin \frac{2\pi}{4} \cdot 2 = \cos \pi + i \sin \pi = -1$$

$$w_4^3 = \dots = -i$$

$$w_4^4 = \dots = w_4^0 = 1$$



$$w_n^{n-k} = w_n^n \cdot w_n^{-k} = w_n^{-k}$$

Hilbert space (finite) $\cong$ vector space
dim space

$$\dim H = n \quad H \cong \mathbb{C}^n$$

$$\langle \cdot | \cdot \rangle = \text{inner product} : H \times H \rightarrow \mathbb{C}$$

$$\langle v | v \rangle \geq 0, \quad \langle v | v \rangle = 0 \Leftrightarrow v = 0$$

$$\langle u | w \rangle = \langle w | u \rangle^* \quad \text{conjugate}$$

$$\langle v | \alpha w + \beta z \rangle = \alpha \langle v | w \rangle + \beta \langle v | z \rangle$$

$$\langle v | w \rangle = \sqrt{\sum_{s=0}^n v_s \cdot w_s}$$

$$\|v\|_2 = \sqrt{\langle v | v \rangle} = \sqrt{\sum_{s=0}^n |v_s|^2}$$

$$|v\rangle = \text{ket-}v = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\langle v| = \text{bra-}v = [\alpha^*, \beta^*]$$

$$\langle v | w \rangle = \text{bra-ket} = \text{scalar}$$

$$|w\rangle\langle v| = \text{rank-1 matrix}$$

$$|v\rangle = \begin{bmatrix} 0.4 \\ 0.8 \\ -0.4 \\ -0.4i \end{bmatrix}$$

$$\langle v| = [0.4, 0.8, -0.4, 0.2i]$$

$$\|v\| = \sqrt{0.16 + 0.64 + 0.16 + 0.16} = 1$$

$$P = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow i$$

Def:

quantum state space is a complex vector space

qubit = complex vector 2-dim

Def: quantum state is a unit-vector (1-dim) in a quantum state space

$$\|v\| = 1$$

$$|v\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C}$$

Computational basis

$n=2$

ground state

extend base

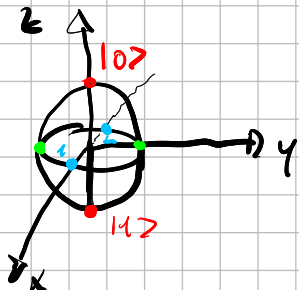
$$|0\rangle \quad |1\rangle$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|v\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$\alpha, \beta \in \mathbb{C}$$

$$|\alpha|^2 + |\beta|^2 = 1$$



$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|+\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

$$G_z |0\rangle = |0\rangle$$

$$G_z |+\rangle = |+\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \rightarrow H |0\rangle = |+\rangle$$

$$H |1\rangle = |-\rangle$$

$$H^2 = I$$

$$H |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = e^{i\theta} \frac{1}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\frac{1+i}{1-i} = -i$$

Rotation of qubits on the Bloch-sphere

$$X = G_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = G_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$Z = G_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{Pauli matrices}$$

$$G_x |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$G_x |1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$G_x |+\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$G_x |-\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = -|-\rangle$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad G_i G_j = -G_j G_i \quad \forall i, j \in \{x, y, z\}$$

$$G_i^2 = I \quad G_i^{-1} = G_i$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$G_x(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle$$

$$S|0\rangle = |0\rangle$$

$$S|1\rangle = i|1\rangle$$

$$G_y(\alpha|0\rangle + \beta|1\rangle) = \begin{bmatrix} -\beta \\ \alpha \end{bmatrix} = -\beta|0\rangle + \alpha|1\rangle$$

$$S \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix}$$

$$G_z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle$$

$$H(\alpha|0\rangle + \beta|1\rangle) = \alpha|+\rangle + \beta|-\rangle$$

$$S(\alpha|0\rangle + i\beta|1\rangle)$$

$$T(\alpha|0\rangle + \beta e^{i\pi/4}|1\rangle)$$

UNITARY MATRIX

$$|V\rangle : \|V\| = 1 \Rightarrow |Y\rangle = U|V\rangle : \|Y\| = 1$$

n dim quantum state space $\cong \mathbb{C}^n$

$$|0\rangle, |1\rangle, \dots, |n-1\rangle \rightarrow \text{unit-vectors}$$

$$U = \begin{bmatrix} | & & | \\ c_0 & & c_{n-1} \end{bmatrix} \quad U \in \mathbb{C}^{n \times n}$$

$$U|k\rangle = |c_k\rangle$$

(*) le colonne di U (unit-matrix) sono unit-vec

$$U \left(\frac{|k\rangle + |5\rangle}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} U|k\rangle + \frac{1}{\sqrt{2}} U|5\rangle = \frac{1}{\sqrt{2}} (|c_k\rangle + |c_5\rangle)$$

$$1 = \left\| \frac{|c_k\rangle + |c_s\rangle}{\sqrt{2}} \right\|^2 = \underbrace{\left(\frac{1}{\sqrt{2}} \right)}_u \cdot \underbrace{\left(\frac{1}{\sqrt{2}} \right)}_u =$$

$$\frac{1}{2} \left(\underbrace{\langle c_k | c_k \rangle}_1 + \underbrace{\langle c_k | c_s \rangle}_u + \underbrace{\langle c_s | c_k \rangle}_u + \underbrace{\langle c_s | c_s \rangle}_1 \right)$$

$$\langle c_s | c_k \rangle + \langle c_k | c_s \rangle = 0$$

$$\begin{array}{ccc} \downarrow & \underbrace{\langle c_s | c_k \rangle}_u & \\ a+ib & \downarrow & \\ \cancel{a+ib} & = 0 & \Rightarrow \operatorname{Re}(\langle c_s | c_k \rangle) = 0 \end{array}$$

$$U \left(\frac{|k\rangle + i|s\rangle}{\sqrt{2}} \right) = \frac{|c_k\rangle + i|c_s\rangle}{\sqrt{2}}$$

$$\operatorname{Im}(\langle c_s | c_k \rangle) = 0$$

$$\langle c_k | c_s \rangle = 0$$

② le colonne di U sono
ortogonali

① + ② $\Rightarrow$ colonne di U ortonormali